

Лабораторная работа №6. Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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2026-03-20

1. Цель работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

2. Порядок выполнения работы и результаты

2.1 Определение полного имени домашнего каталога

```
(base) hhl@LAPTOP-058L8DAA:~$ pwd  
/home/hhl
```

```
pwd
```

Результат:

```
/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
```

2.2 Работа с каталогом /tmp

2.2.1 Переход в каталог

```
(base) hhl@LAPTOP-058L8DAA:~$ cd /tmp
```

```
cd /tmp  
pwd
```

2.2.2 Просмотр содержимого

```
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls
snap-private-tmp
systemd-private-07c39e09d0c340849ball14be17flacc0-polkit.service-k0DwV5
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-logind.service-2VhshC
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -l
total 24
drwx----- 2 root root 4096 Mar 19 00:05 snap-private-tmp
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-polkit.service-k0DwV5
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-logind.service-2VhshC
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-resolved.service-Cpw3M5
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-timesyncd.service-sy2BDZ
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-ssl-pro.service-IyV6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -a
.
.
snap-private-tmp
systemd-private-07c39e09d0c340849ball14be17flacc0-polkit.service-k0DwV5
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-logind.service-2VhshC
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-resolved.service-Cpw3M5
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-timesyncd.service-sy2BDZ
systemd-private-07c39e09d0c340849ball14be17flacc0-ssl-pro.service-IyV6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -la
total 40
drwxrwxrwt 9 root root 12288 Mar 19 00:06 .
drwxr-xr-x 22 root root 4096 Mar 19 00:05 ..
drwxrwxrwx 2 root root 60 Mar 19 00:05 .X11-unix
drwx----- 2 root root 4096 Mar 19 00:05 snap-private-tmp
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-polkit.service-k0DwV5
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-logind.service-2VhshC
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-resolved.service-Cpw3M5
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-timesyncd.service-sy2BDZ
drwx----- 3 root root 4096 Mar 19 00:05 systemd-private-07c39e09d0c340849ball14be17flacc0-ssl-pro.service-IyV6Lo
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls -F
snap-private-tmp/
systemd-private-07c39e09d0c340849ball14be17flacc0-polkit.service-k0DwV5/
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-logind.service-2VhshC/
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-resolved.service-Cpw3M5/
systemd-private-07c39e09d0c340849ball14be17flacc0-systemd-timesyncd.service-sy2BDZ/
systemd-private-07c39e09d0c340849ball14be17flacc0-ssl-pro.service-IyV6Lo/
```

```
ls
ls -l
ls -a
ls -la
ls -F
```

Пояснение:

Команда	Описание
ls	Список файлов
ls -l	Подробный список
ls -a	Скрытые файлы
ls -la	Всё + подробно
ls -F	Типы файлов

2.2.3 Проверка cron

```
(base) hhl@LAPTOP-058L8DAA:/tmp$ ls /var/spool
cron  libreoffice  mail  rsyslog
```

```
ls /var/spool
```

2.2.4 Возврат домой

```
(base) hhl@LAPTOP-058L8DAA:~/tmp$ cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ pwd
/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls -l
total 8
drwxr-xr-x 4 hhl hhl 4096 Feb 22 04:03 presentation
drwxr-xr-x 6 hhl hhl 4096 Feb 22 04:03 report
```

```
cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06
pwd
ls -l
```

2.3 Работа с каталогами

Создание каталогов

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir newdir
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls -ld newdir
drwxr-xr-x 2 hhl hhl 4096 Mar 19 00:15 newdir
```

```
mkdir newdir
ls -ld newdir
```

Подкаталог

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir newdir/morefun
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls newdir
morefun
```

```
mkdir newdir/morefun
ls newdir
```

Несколько каталогов

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ mkdir letters memos misk
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls
letters memos misk newdir presentation report
```

```
mkdir letters memos misk
ls
```

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rmdir letters memos misk
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls
newdir presentation report
```

```
rmdir letters memos misk
ls
```

Ошибка rm

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rm newdir
rm: cannot remove 'newdir': Is a directory
```

```
rm newdir
```

Ошибка:

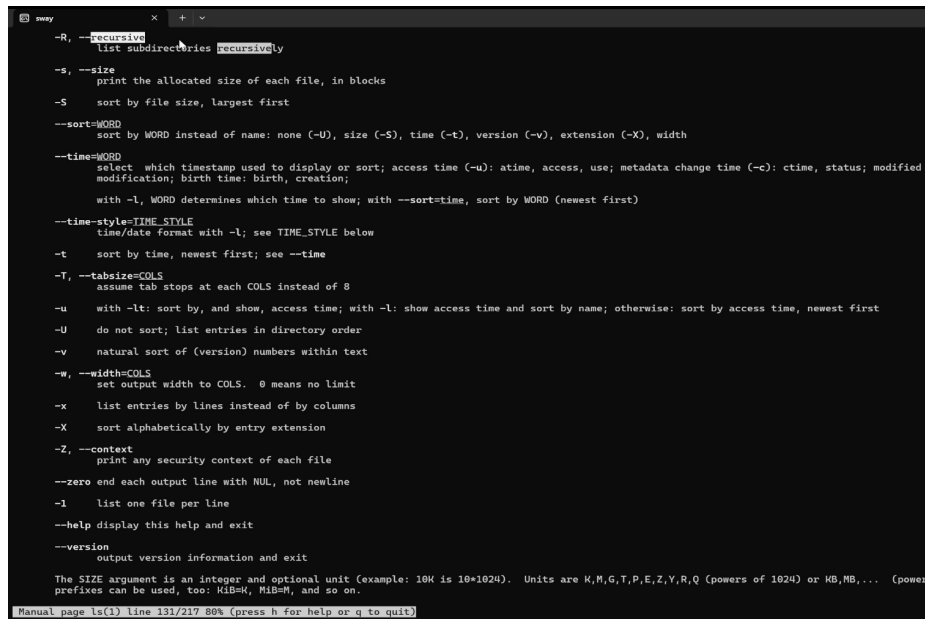
```
rm: cannot remove 'newdir': Is a directory
```

Удаление подкаталога

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ rmdir newdir/morefun
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ ls newfir
ls: cannot access 'newfir': No such file or directory
```

```
rmdir newdir/morefun
ls newdir
```

2.4 Рекурсивный вывод



```
-R, --recursive
    list subdirectories recursively

-s, --size
    print the allocated size of each file, in blocks

-S
    sort by file size, largest first

--sort=WORD
    sort by WORD instead of name (-U), size (-S), time (-t), version (-v), extension (-X), width

--time=WORD
    select which timestamp used to display or sort; access time (-u): atime, access, use; metadata change time (-c): ctime, status; modified
    modification; birth time: birth, creation;
    with -l, WORD determines which time to show; with --sort=time, sort by WORD (newest first)

--time-style=TIME_STYLE
    time/date format with -l; see TIME_STYLE below

-t
    sort by time, newest first; see --time

-T, --tabsize=COLS
    assume tab stops at each COLS instead of 8

-u
    with -lt: sort by, and show, access time; with -l: show access time and sort by name; otherwise: sort by access time, newest first

-U
    do not sort; list entries in directory order

-v
    natural sort of (version) numbers within text

-w, --width=COLS
    set output width to COLS. 0 means no limit

-x
    list entries by lines instead of by columns

-X
    sort alphabetically by entry extension

-Z, --context
    print any security context of each file

--zero
    end each output line with NUL, not newline

-l
    list one file per line

--help
    display this help and exit

--version
    output version information and exit

The SIZE argument is an integer and optional unit (example: 10M is 10*1024). Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers
of 1000). Prefixes k, m, g, t, p, e, z, y, r, q are also accepted.

Manual page ls(1) line 131/219 80% (press h for help or q to quit)
```

```
ls -R
```

2.5 Сортировка по времени

```
man ls
-t      sort by time, newest first; see --time
-T, --tabsize=COLS
      assume tab stops at each COLS instead of 8
-u      with -lt: sort by, and show, access time; with -l: show access time and sort by name; otherwise: sort by access time, newest first
-U      do not sort; list entries in directory order
-v      natural sort of (version) numbers within text
-w, --width=COLS
      set output width to COLS. 0 means no limit
-x      list entries by lines instead of by columns
-X      sort alphabetically by entry extension
-Z, --context
      print any security context of each file
--zero
      end each output line with NUL, not newline
-l      list one file per line
--help
      display this help and exit
--version
      output version information and exit

The SIZE argument is an integer and optional unit (example: 10K is 10*1024). Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers
of 1000). Prefixes can be used, too: MiB=M, GiB=G, and so on.

The TIME_STYLE argument can be full-iso, long-iso, iso, locale, or +FORMAT. FORMAT is interpreted like in date(1). If FORMAT is FORMAT1-newline+
FORMAT2, FORMAT1 applies to non-recent files and FORMAT2 to recent files. TIME_STYLE prefixed with 'posix-' takes effect only outside the POSIX locale. Also th
e TIME_STYLE environment variable sets the default style to use.

The WHEN argument defaults to 'always' and can also be 'auto' or 'never'.

Using color to distinguish file types is disabled both by default and with --color=never. With --color=auto, ls emits color codes only when stand
ardized by a terminal. The LS_COLORS environment variable can change the settings. Use the dircolors(1) command to set it.

Exit status:
 0      if OK,
 1      if minor problems (e.g., cannot access subdirectory),
 2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR  Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
  Manual page ls(1) line 151/217 92% (press h for help or q to quit)
```

```
ls -lt
ls -ltr
```

2.6 Изучение команд

cd

```
(base) hhi@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ man cd

cd ~
cd ..
cd -
```

pwd

```

PWD(1) User Commands
NAME
    pwd - print name of current/working directory
SYNOPSIS
    pwd [OPTION]...
DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        avoid all symlinks

    --help
        display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    NOTE: your shell may have its own version of pwd, which usually supersedes the version described here. Please refer to your shell's documentation for the options it supports.
AUTHOR
    Written by Jim Meyer.
REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>
COPYRIGHT
    Copyright © 2023 Free Software Foundation, Inc. License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
    This is free software: you are free to change and redistribute it. There is NO WARRANTY, to the extent permitted by law.
SEE ALSO
    getcwd(3)

    Full documentation <https://www.gnu.org/software/coreutils/pwd>
    or available locally via: info '(coreutils) pwd invocation'
GNU coreutils 9.4 April 2024
Manual page pwd(1) line 1/45 (END) (press h for help or q to quit)

```

```

pwd -L
pwd -P

```

mkdir

```

MKDIR(1) User Commands
NAME
    mkdir - make directories
SYNOPSIS
    mkdir [OPTION]... DIRECTORY...
DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected by any -m option.

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help
        display this help and exit

    --version
        output version information and exit
AUTHOR
    Written by David MacKenzie.
REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>
COPYRIGHT
    Copyright © 2023 Free Software Foundation, Inc. License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
    This is free software: you are free to change and redistribute it. There is NO WARRANTY, to the extent permitted by law.
SEE ALSO
    mkdir(2)

    Full documentation <https://www.gnu.org/software/coreutils/mkdir>
    or available locally via: info '(coreutils) mkdir invocation'
GNU coreutils 9.4 April 2024
/-#

```

```

mkdir -p dir1/dir2/dir3
mkdir -m 755 dir

```

rmdir

```
RMDIR(1) User Commands
NAME
    rmdir - remove empty directories
SYNOPSIS
    rmdir [OPTION]... DIRECTORY...
DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.
    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory
    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'
    -v, --verbose
        output a diagnostic for every directory processed
    --help
        display this help and exit
    --version
        output version information and exit
AUTHOR
    Written by David MacKenzie.
REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>
COPYRIGHT
    Copyright © 2023 Free Software Foundation, Inc. License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
    This is free software: you are free to change and redistribute it. There is NO WARRANTY, to the extent permitted by law.
SEE ALSO
    rmdir(2)
    Full documentation <https://www.gnu.org/software/coreutils/rmdir>
    or available locally via: info '(coreutils) rmdir invocation'
GNU coreutils 9.4 April 2024
Manual page rmdir(1) line 1/43 (END) (press h for help or q to quit)
```

```
rmdir -p dir1/dir2/dir3
```

rm

```
RM(1) User Commands
NAME
    rm - remove files or directories
SYNOPSIS
    rm [OPTION]... [FILE]...
DESCRIPTION
    This manual page documents the GNU version of rm. rm removes each specified file. By default, it does not remove directories.
    If the -i or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the
    user to proceed with the entire operation. If the response is not affirmative, the entire command is aborted.
    Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always o
    prompts the user for whether to remove the file. If the response is not affirmative, the file is skipped.
OPTIONS
    Remove (unlink) the FILE(s).
    -f, --force
        ignore nonexistent files and arguments, never prompt
    -i
        prompt before every removal
    -I
        prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection aga
    --interactive[=WHEN]
        prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always
    --one-file-system
        when removing a hierarchy recursively, skip any directory that is on a file system different from that of the corresponding command line ar
    --no-preserve-root
        do not treat '/' specially
    --preserve-root[=all]
        do not remove '/' (default); with 'all', reject any command line argument on a separate device from its parent
    -r, -R, --recursive
        remove directories and their contents recursively
    -d, --dir
        remove empty directories
    -v, --verbose
        explain what is being done
    --help
        display this help and exit
    --version
        output version information and exit
Manual page rm(1) line 1 (press h for help or q to quit)
```

```
rm file.txt
rm -r dir
rm -rf dir
```

2.7 История команд

```
1864 quarto render report.qmd
1865 cd ~/work/study/2026-2027/Операционные\ системы/os-intro/labs/lab05
1866 gedit report.qmd
1867 quarto render report.qmd
1868 gedit report.qmd
1869 gedit presentation.qmd
1870 quarto render presentation.qmd
1871 conda activate sage
1872 sgae
1873 sage
1874 conda activate sage
1875 sage
1876 conda activate sage
1877 sage
1878 cd ~/work/study/2026-2027/Операционные\ системы/os-intro/labs/lab06
1879 pwd
1880 cd /tmp
1881 cd
1882 cd /tmp
1883 cd /tmp
1884 cd
1885 cd ~/work/study/2026-2027/Операционные\ системы/os-intro/labs/lab06
1886 cd /tmp
1887 pwd
1888 cd /tmp
1889 ls
1890 ls -l
1891 ls -a
1892 ls -la
1893 ls -F
1894 ls /var/spool
1895 cd ~/work/study/2026-2027/Операционные\ системы/os-intro/labs/lab06
1896 pwd
1897 ls -l
1898 mkdir newdir
1899 ls -ld newdir
1900 mkdir newdir/morefun
1901 ls newdir
1902 mkdir letters memos misk
1903 ls
1904 rmdir letters memos misk
1905 ls
1906 rm newdir
1907 ls -ld newdir
1908 rmdir newdir/morefun
1909 ls newfir
1910 man ls
1911 man cd
1912 man pwd
1913 man mkdir
1914 man rmdir
1915 man rm
1916 history
```

history

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab06$ !1891:s/a/F
ls -F
newdir/ presentation/ report/
```

!1891
!1891:s/a/F

3. Ответы на вопросы

Командная строка — текстовый интерфейс управления системой.

Определение пути:

pwd

Типы файлов:

ls -F

Скрытые файлы:


```
ls -a  
ls -la
```

Удаление:

```
rm file  
rmdir dir  
rm -r dir
```

История:

```
history
```

Модификация:

```
!3:s/a/F
```

Несколько команд:

```
cd /tmp; ls  
mkdir dir && cd dir  
cd bad || echo "error"
```

Экранирование:

```
touch Мой\ файл.txt  
echo "Цена \$100"
```

ls -l: Права, владелец, размер, дата, имя.

Пути:

```
cd /absolute/path  
cd relative/path
```

Справка:

```
man ls  
ls --help
```

Автодополнение: клавиша Tab

4. Выводы

- Освоены команды cd, pwd
- Изучен ls
- Освоены mkdir, rm, rmdir
- Работа с man
- Изучена история команд
- Освоены пути и экранирование

Все задачи выполнены.